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Key Points:

- Three-dimensional simulations of reactive transport processes using the micro-continuum approach based on a fractured dolostone experiment
- The altered layer (AL) as a diffusion barrier limits reactions of the fast-reacting mineral and thus the matrix alteration
- Advection in the AL affects spatial patterns of mineral dissolution that controls subsequent AL development and fracture enlargement

Supporting Information:

Supporting Information may be found in the online version of this article.

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Investigation of Coupled Processes in Fractures and the Bordering Matrix via a Micro-Continuum Reactive Transport Model

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Abstract In multi-mineral fractured rocks, the altered porous layer on the fracture surface resulting from preferential dissolution of the fast-reacting minerals can have profound impacts on subsequent chemical-physical alteration of the fractures. This study adopts the micro-continuum approach to provide further understanding of reactive transport processes in the altered layer (AL), and mass exchanges with the bordering matrix and fracture. The modeling framework couples the Darcy-Brinkman-Stokes (DBS) solver in COMSOL Multiphysics and the geochemical modeling capability of CrunchFlow. Three-dimensional steady state simulations with systematically varied chemical-physical parameters of the AL were performed to examine the impacts of individual factors and processes. Our simulation results confirm previous observations that dissolution of the fast-reacting mineral (i.e., calcite) is largely controlled by diffusion across the AL. We also show that dissolution of the slow-reacting mineral (i.e., dolomite), which controls AL development and fracture enlargement, increases with surface area and has a complex dependence on different local rate-limiting processes. In particular, advection can result in evident spatial variations in the local dissolution rates of dolomite, although it does not affect the bulk chemistry significantly. The difference in the spatial patterns between simulations with and without advection in the AL is more noticeable in the locations with smaller apertures, with up to 20% difference in local reaction rates. Therefore, it is important to include a full depiction of advection, diffusion, and reactions for accurately capturing local dynamics that control long-term fracture evolution.

1. Introduction

Fractures act as preferential flow pathways that govern fluid migration and solute transport in many natural and engineered systems, and are subject to alteration driven by geochemical reactions (Berkowitz et al., 2016; Deng & Spycher, 2019; Steefel, Beckingham, & Landrot, 2015). Large contrasts in morphological and mineralogical properties between fractures and bordering rock media can result in non-linear feedback between reactions, fluid flow, and transport (Molins et al., 2017; Schädle et al., 2019). Improved fundamental understanding of these coupled processes in fractured rocks is critical for accurate assessment of the long-term performance of different energy and environmental systems (Mehmani et al., 2020; Noiriel & Daval, 2017).

Experimental observations and numerical simulations have explored the alteration of fracture morphology and thus hydrophysical properties induced by changes in fluid composition, temperature, and pressure (Al-Khulaifi et al., 2017; Davila et al., 2016; Deng & Spycher, 2019). Dimensionless numbers, including Péclet and Damköhler numbers, have been used to investigate and predict dissolution behaviors in different flow regimes (Brunet et al., 2016; Detwiler et al., 2003; Szymczak & Ladd, 2009). These studies, however, typically focus on single-mineral systems in which mineral dissolution/precipitation mainly occur at the fracture surface, whereas mineral heterogeneity that is widely observed can lead to complex alteration in the properties of the bordering rock matrix (Upadhyay et al., 2015). In multi-mineral systems with varied reactivities, preferential dissolution of the fast-reacting minerals results in the development of an altered porous layer bordering the fracture, which affects the subsequent reactions (Laubach et al., 2019; Wen et al., 2016), and thus the evolution of fracture morphology and hydraulic conductivity (Ellis et al., 2011; Noiriel et al., 2013).

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Continuum scale models have been developed to demonstrate and quantify the role of the altered layer (AL) as a diffusion barrier, which limits mass transfer between the fracture and rock matrix (Deng et al., 2016; Deng, Molins, et al., 2018; Noiriel & Daval, 2017). While these modeling studies have enabled predictions of AL development and their impacts on reactions using simplified treatments, they are limited to specific conditions and they only partially represent the coupled (flow-)transport-reaction processes in the AL. For instance, while the effective diffusion coefficient that was introduced to describe the transport limitation on reaction rate of the fast-reacting minerals (Deng et al., 2016) can potentially be adjusted to account for the effects of advection in the AL, it would be purely empirical unless physically based understanding of such effects is taken into consideration (Scheibe et al., 2015). In this regard, the relative importance of different processes within the AL in controlling the overall fracture evolution is still an open question.

Recent advances in microscopic characterization techniques and computational power have enabled considerations of reactive transport processes in heterogeneous fractured porous media at the pore-scale (Deng et al., 2020; Molins, 2015; Singh et al., 2017). In pore-scale models, such as the mesh based pore-scale reactive transport model (Bultreys et al., 2016; Molins et al., 2014; Starchenko & Ladd, 2018), lattice Boltzmann method (LBM; Curti et al., 2019; Kang et al., 2010; Prasianakis et al., 2017), and smooth particle hydrodynamics (SPH; Tartakovsky et al., 2007, 2016), the interfaces between fluid and solid phases are explicitly resolved and the flow field reflects fine-scale morphology. Chemical gradients arise from the interplay between local advective transport, diffusion, and chemical reactions can be evaluated to reveal surfaces versus transport limitations on reactivity (Deng, Steefel, Molins, & DePaolo, 2018; Fazeli et al., 2019). As such, pore-scale models can be a valuable tool to connect microscopic geometric heterogeneity and the macroscopic properties (e.g., permeability and reaction rate; Molins & Knabner, 2019; Starchenko & Ladd, 2018).

However, the morphological and mineralogical heterogeneity in the fracture and the rock matrix can vary across a wide range of scales, resulting in a scale dependence of coupled processes and parameters (Al-Khulaifi et al., 2019; Jung & Navarre-Sitchler, 2018). In addition to the coupling of multiple processes, the controlling processes with different characteristic length scales and their interactions need to be identified and accounted for. Pore-scale simulations have been performed in two-dimensional regions or in a synthetic binary mineral system to investigate mass transfer between fractures and altering rock matrix, they, however, did not consider the dissolution of slow-reacting minerals (Chen et al., 2014; Fazeli et al., 2019). The application of pore-scale models in three-dimensional (3D) cm-scale experimental or natural systems is even more rare.

In light of the scale differences and computational challenges of the pore-scale approach, reactive transport modeling in fractured media can be conceptualized as a multiscale problem that couples a pore-scale component in the fracture, and a Darcy-scale component in the porous matrix (Molins et al., 2019), which is commonly referred to as multiscale modeling (Molins & Knabner, 2019; Yang et al., 2021). One approach of multiscale modeling involves the use of hybrid algorithms, which divide the domain into subdomains according to their respective characteristic length scales. Different governing equations are solved for each subdomain and the subdomains are coupled by enforcing mass and momentum continuity at the interface (Mehmani & Balhoff, 2015; Tartakovsky et al., 2008; Tomin & Lunati, 2013; Yousefzadeh & Battiato, 2017). However, implementation of different discretization across the interface can be complicated, especially when the interface moves as a result of reactions, as different resolutions are required to capture features in each individual subdomain (Molins & Knabner, 2019).

Another multiscale modeling approach involves the implementation of the Darcy-Brinkman-Stokes (DBS) equation in a single micro-continuum domain (Carrillo et al., 2020; Gulbransen et al., 2010; Soulaine et al., 2017; Soulaine & Tchelepi, 2016). The DBS equation solves for the Stokes (or Navier-Stokes) flow and Darcy flow simultaneously and has been used across spatial scales under changing flow regimes (Brinkman, 1947; Neale & Nader, 1974). Reactive transport models using the DBS approach have been developed for various applications. Golfier et al. (2002) applied a DBS-based model at Darcy-scale to capture the wormhole development of different dissolution regimes. Soulaine and Tchelepi (2016) proposed a DBS micro-continuum model for pore-scale simulations of evolving porous media. Although the fluid-solid interfaces are not explicitly resolved as in the pore-scale models, the micro-continuum approach captured pore structure evolution and has the advantage of accommodating complex geometries without requiring extremely high resolution or adding significant computational costs. Compared to the hybrid methods, the micro-continuum approach unites the mathematical description of features at different scales using one set of governing equations and enables computationally efficient representation of heterogeneity and all relevant processes at both pore- and continuum scale. It is especially useful

10.1029/2021WR030578

to capture the reactive transport processes through porous regions and solid-free regions (Carrillo et al., 2020; Molins et al., 2021). In spite of these advantages, this approach has not been applied to simulate reactive transport in fractured media, a prime example of multiscale media in that there is a sharp contrast—with a clear separation of scales—between the porosity in the fracture opening and in the rock matrix (Deng & Spycher, 2019).

Our study focuses on reactive transport processes in fractured media, particularly in the porous AL bordering the fracture and their impacts on fractured alteration. For this purpose, we perform 3D multiscale simulations of a fractured porous medium with a spatially heterogeneous AL bordering the fracture using the micro-continuum approach, including a full representation of advection, diffusion, and reactions (dissolution of both the fast-reacting mineral and the slow-reacting mineral) in the entire domain. This is enabled by a framework that couples a multiphysics code (COMSOL Multiphysics, COMSOL) with a reactive transport code (CrunchFlow). The mathematical model and numerical implementation of this coupling framework are described in Section 2. In Section 3, we present the setup of the 3D simulations, and we report the results and discuss the impacts of different physical-chemical parameters of the AL on the reactive transport dynamics in the fractured multiscale domain in Section 4. We conclude in Section 5.

2. Materials and Methods

2.1. Governing Equations

The micro-continuum approach is derived by volume averaging governing equations over each control volume (V; Beavers & Joseph, 1967; Beckermann & Viskanta, 1988) instead of a full pore-scale representation. The control volumes may contain both a solid and a fluid phase or one of them. The volume fraction of fluid (ε_f) is 0 and 1 in the cells containing only solid or fluid, respectively. Cells with intermediate values ($0 < \varepsilon_f < 1$) are identified as the fluid/solid interface when bordering the fluid cells, and are otherwise part of the porous matrix. In each cell or control volume, all the physical variables are volume averages.

2.1.1. Flow

Flow in a medium composed of porous and non-porous regions can be described by the single field momentum conservation equation arising from the integration of the Navier-Stokes (N-S) momentum equation (Bennon & Incropera, 1987; Bousquet-Melou et al., 2002; Brinkman, 1947; Ochoa-Tapia & Whitaker, 1995; Quintard & Whitaker, 1999):

$$\frac{1}{\varepsilon_f}\left(\rho_f\frac{\partial\bar{\mathbf{u}}_f}{\partial t}+\rho_f\bar{\mathbf{u}}_f\cdot\nabla\left(\frac{\bar{\mathbf{u}}_f}{\varepsilon_f}\right)\right)=-\nabla\bar{p}_f+\frac{\mu_f}{\varepsilon_f}\nabla^2\bar{\mathbf{u}}_f-\frac{\mu_f}{\mathbf{k}}\bar{\mathbf{u}}_f \tag{1}$$

where $\mathbf{k}$ is the permeability tensor of the medium, and the velocity is defined as

$$\bar{\mathbf{u}}_f=\frac{1}{V}\int_{V_f}\mathbf{u}_f\,dV, \tag{2}$$

physical properties of the fluid, that is, fluid density (ρ_f) and fluid viscosity (μ_f) and the pressure field are defined as intrinsic volume averages that correspond to the volume occupied by the fluid (V_f).

The left-hand side in Equation 1 describes the inertia, and the right-hand side terms represent the pressure gradient, the diffusive viscous contribution, the momentum exchange, respectively. The momentum exchange subject to the mutual friction between fluids and solids vanishes in solid-free regions (large k values) while tends to be dominant in porous regions where permeability is low. As such, this equation enables us to recover the NS flow field in the pore space when local porosity and permeability fields in the solid phase are set to arbitrarily small values (Molins et al., 2021). Moreover, Soulaine and Tchelepi (2016) demonstrated that no-slip boundary conditions at the solid/fluid interfaces can be ensured when k within the mineral is at least four orders of magnitude below what is in the fluid zone.

2.1.2. Reactive Solute Transport

Reactive transport is solved by the averaged multicomponent advection-diffusion-reaction equation:

$$\frac{\partial(\varepsilon_f\bar{\psi}_i)}{\partial t}=-\nabla\cdot(\bar{\mathbf{u}}_f\varepsilon_f\bar{\psi}_i)+\nabla\cdot(\varepsilon_f\mathbf{D}_i\nabla\bar{\psi}_i)+R_i \tag{3}$$

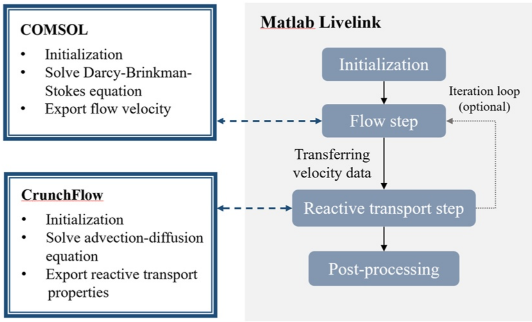

Figure 1. Schematic diagram of the structure of the COMSOL-CrunchFlow framework.

where $\bar{\psi}_i$ is the molal total concentration of a component, $\mathbf{D}_i$ is the combined dispersion-diffusion tensor, and R_i is the changes in the concentration of the species due to mineral reactions. For a multi-mineral system, $R_i = \sum_s \upsilon_{i,s} r_s$, where $\upsilon_{i,s}$ is the stoichiometric coefficient of reaction s, and the mineral reaction rate r_s for a given mineral is given by the transition state theory rate law (Lasaga, 2018):

$$r_s = -A_s \left\{ \sum_{l=1}^{N_r} k_l \left(\prod_{i=1}^{N_c+N_x} a_i^{p_{il}} \right) \right\} \left[1 - \left(\frac{Q_s}{K_s} \right)^M \right]^n, \tag{4}$$

where A_s refers to the surface area of individual minerals, k_l is the far from equilibrium dissolution rate constant for the lth parallel reaction, Q_s is the ion activity product, and K_s is the equilibrium constant. The term $\prod_{i=1}^{N_c+N_x} a_i^{p_{il}}$ incorporates the effects of catalytic/inhibitory ions in the solution. The exponents n and M allow for non-linear dependencies on the saturation state and are typically fitted from experimental studies. Both are one in our simulations.

2.2. Numerical Implementation

This framework is implemented by coupling the COMSOL, that is, a multi-physics and multi-purpose software, with CrunchFlow, that is, a software package with comprehensive geochemical modeling capability (Steefel, Appelo, et al., 2015; Steefel & Lasaga, 1994). The DBS equation for single phase flow (Equation 1) is solved in COMSOL, and the advection-diffusion-reaction equation is solved in CrunchFlow for the transport of the reactive species and the resulting evolution of pore structure when geometry update is considered.

The program structure of this coupling framework is illustrated in Figure 1. At the beginning of the simulation, domain geometry, discretization, properties, boundaries are initialized in COMSOL and CrunchFlow separately. After the initialization procedure, Matlab LiveLink is used as the application interface to step through COMSOL and CrunchFlow for the calculation of flow and reactive transport, respectively. All data, that is, fluid chemistry, porosity, and permeability, are stored and updated in CrunchFlow. The flow model in COMSOL discretizes the domain into an unstructured mesh, and computed using the finite element method. The velocity field is then mapped to the Cartesian mesh using the module of data export in COMSOL and transferred into CrunchFlow. The mesh resolution is presented later in the simulation setup section. In this study, we focus on steady state simulations, and thus no iteration between COMSOL and CrunchFlow is needed. But the framework can be used for cases with geometry evolution by adding the iteration loops. Both COMSOL and CrunchFlow have been extensively validated against known solutions and benchmark problems (Azad et al., 2016; Molins et al., 2021; Nardi et al., 2014)

3. Setup of the Multiscale Model in 3D Fractured Dolostone Experiment

A 3D multi-scale model is built using the coupled COMSOL-CrunchFlow framework based on a previous fracture-flow experiment to investigate the reactive transport processes in a fractured medium with complex geometries and mineralogy. In the experiment, CO_2-saturated brine was injected into a fractured Duperow Dolostone, which resulted in the formation of a preferential channel and the development of an AL (Ajo-Franklin et al., 2018). This experiment has served as a test case for several modeling studies, and has been simulated by Deng et al. (2016) using a reduced-dimension Darcy-scale model and by Molins et al. (2019) using a hybrid reactive transport model in which flow and reactions were considered in the fracture domain and diffusion and reactions were included in the matrix domain. Here, similar to Molins et al. (2019), we simulate the steady state of a time snapshot from the experiment, but with full coupling of flow, transport, and reactions in both the fractures and the altered/unreacted matrix.

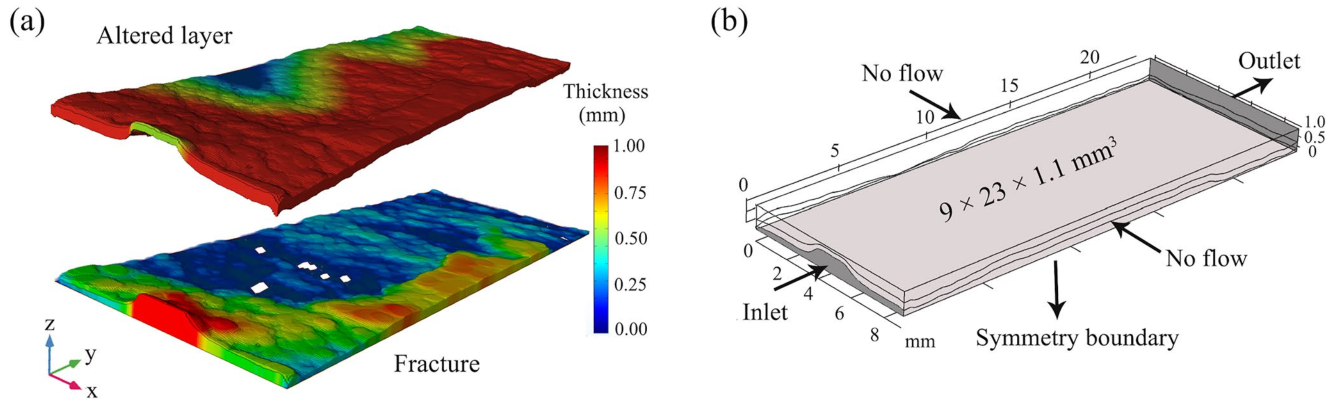

Figure 2. The (a) 3D domains and (b) boundary conditions for multi-scale models. The altered layer and the fracture were reconstructed from the thickness and aperture maps, respectively (fracture contact points are displayed in white). The domain bottom, that is, the central plane of the fracture was set as a symmetry boundary.

3.1. Multi-Scale Simulation Setup

The 3D domain is reconstructed from the fracture aperture and AL thickness maps at 80 hr from Deng et al. (2016). Assuming the core is symmetric with respect to the fracture plane, only half of the domain (Figure 2) was used for simulation. The aperture of the fracture ranges from 0 to 0.805 mm, and the thickness of the AL ranges from 0.028 to 1.009 mm. The entire domain is $9 \times 23 \times 1.1$ mm^3 in size.

The multi-scale medium includes three zones, that is, the open fracture region, the AL characterized by the absence of calcite, and the unreacted rock matrix, with a volume fraction of 15.68%, 24.88%, and 59.44%, respectively. At the inlet, the reactive fluid was injected into the fracture with a constant rate of 2.5 μL/min (half of the experimental rate because half of the domain was simulated), and a constant pressure was used at the outlet. The symmetry boundary condition was applied at the bottom surface of the domain representing the central plane of the fracture, and other boundaries are without flow or mass transfer. The flow model in COMSOL used an unstructured mesh with 4.26e6 elements and the finite element method to solve for the steady state velocity field. The velocity was mapped to the Cartesian mesh ($60 \times 200 \times 110$, 1.32e6 cells) used in CrunchFlow to solve the advection-diffusion-reaction equation using the integrated finite difference method. The resolution of the CrunchFlow mesh used in this work is similar to the initial setup of the 3D hybrid model (Molins et al., 2019), which has $64 \times 160 \times 32$ cells. The vertical resolution in the hybrid model is coarser because adaptive mesh refinement was implemented. In our model, the resolution is 10 μm for the z-axis, which is close to the data resolution of the fracture aperture and thus able to capture the roughness. A mesh convergence study was performed in a small region—one tenth of the entire domain—to evaluate the effect of mesh resolutions on both flow and reactive transport simulations. Results of the pressure difference, and the effluent Ca and Mg concentrations differ by <0.02% when the number of grid cells increases by a factor of 2. It confirms that the mesh used in our simulations is fine enough. Furthermore, details regarding the mesh convergence study are presented in the Supporting Information. The computational time is 4–6.5 CPU hours for the COMSOL simulations and 100–250 CPU hours for the CrunchFlow simulations depending on the scenarios tested.

Following the experimental characterization, dolomite and calcite were assumed to be well mixed in the unreacted matrix. The volume fraction of dolomite is 81% in both the AL and matrix, and calcite is only present in the unreacted matrix with a volume fraction of 9%. Therefore, the porosity is 1, 0.19, and 0.10 in the fracture, the AL, and the matrix, respectively. For the steady state simulations, the porosities and volume fractions were not updated. Initial and boundary conditions are summarized in Table 1. The values of permeability were determined based on permeability measurements at different porosities reported in previous experimental studies (Ehrenberg et al., 2006; Kaminskaite et al., 2019). The rate equations for calcite and dolomite are given by (Chou et al., 1989; Deng et al., 2016) as follows

$$r_{CaCO_3} = (k_1 \gamma_{H^+} C_{H^+} + k_2 \gamma_{H_2CO_3^*} C_{H_2CO_3^*} + k_3) \left(1 - \frac{Q_{CaCO_3}}{K_{CaCO_3}}\right) \tag{5}$$

Table 1
Chemical and Physical Parameters Used in the Base Case Simulation

Domain	Porosity	Permeability (m^2)	Effective diffusion coefficient (m^2/s)	Reactive surface area (m^2/m^3)	Inlet flow rate (μL/min)	Solution parameters for initial[a]/inlet conditions	
Fracture	Free flow in the fracture zone				2.5	pH	4.8/3.13
Altered layer	0.19	1e−14	2e−10	Dolomite 2,555		pCO_2	1.2/1.2 (mol/kg)
Matrix	0.10	5e−13	4e−12	Calcite 1e6		Mg	5e−3/0 (mol/kg)
				Dolomite 2,555		Ca	2e−2/0 (mol/kg)

[a]Initial conditions were set to be at equilibrium with the mineral phases in order to reach steady state quickly.

$$r_{CaMg(CO_3)_2} = \left(k_4 \gamma_{H^+}^{0.5} C_{H^+}^{0.5}\right)\left(1 - \frac{Q_{CaMg(CO_3)_2}}{K_{CaMg(CO_3)_2}}\right) \tag{6}$$

where K_{CaCO_3}, $K_{CaMg(CO_3)_2}$ are the solubility constants and Q_{CaCO_3}, $Q_{CaMg(CO_3)_2}$ are the ion activity products of the reaction for calcite and dolomite, respectively. C_i and γ_i are the concentrations and activity coefficients of species i. In this single phase system, all aqueous reactions are treated as equilibrium reactions. The kinetic coefficients follow Molins et al. (2019), that is, $k_1 = 0.083$ mol/m²/s, $k_2 = 1.5e{-4}$ mol/m²/s, $k_3 = 1.1e{-4}$ mol/m²/s, and $k_4 = 4.5e{-4}$ mol/m²/s. The effective diffusion coefficients and reactive surface areas are specified following the continuum approach in the porous regions and are also consistent with previous modeling work (Molins et al., 2019).

3.2. Scenario Setup for Analysis of Altered Layer Properties and Processes

Several scenarios were designed for comparison to isolate the impact of flow, transport and reaction in the AL on the spatial patterns of reaction rate and the overall behaviors, and thus to identify the controlling factor on the subsequent evolution. Key parameters of the AL, that is, porosity, permeability, effective diffusion coefficient, and reactive surface area, were varied. Table 2 summarizes the scenarios and the respective input parameters. The DBS_ε0.19 scenario is the base case with the best available parameter values for the experiment and considers all processes. In the DBS_ε0.5 scenario, the AL is made more permeable by increasing the calcite volume fraction and AL porosity to 0.5. N-S_ε0.19 and N-S_ε0.5 scenarios have the same setup as DBS_ε0.19 and DBS_ε0.5,

Table 2
Reactive Transport Parameters for Simulation Scenarios

			Transport metrics				Reaction metrics
		Base case	Advection-diffusion effects	Advection effects[b]		Diffusion effects	Surface area effects
Parameters[a]		DBS_ε0.19 scenario (base case)	DBS_ε0.5 scenario	N-S_ε0.19 scenario	N-S_ε0.5 scenario	DBS_D_{low} scenario	DBS_A_{high} scenario
Porosity_ε	Altered layer	0.19	0.50	0.19	0.50	0.19	0.19
	Matrix	0.10	0.10	0.10	0.10	0.10	0.10
Permeability (m2)[c]	Altered layer	5e−13	**1e−10**	5e−13	**1e−10**	5e−13	5e−13
	Matrix	1e−14	1e−14	1e−14	1e−14	1e−14	1e−14
Effective diffusion coefficient_D (m^2/s)[c]	Altered layer	2e−10	**1e−9**	2e−10	**1e−9**	**4e−11**	2e−10
	Matrix	4e−12	4e−12	4e−12	4e−12	4e−12	4e−12
Surface area_A (m^2/m^3)	Calcite	1e6	1e6	1e6	1e6	1e6	1e6
	Dolomite	2,555	2,555	2,555	2,555	2,555	**12,775**

[a]The bold fonts highlight varied parameters compared to the base case. [b]Velocity fields for N-S_ε0.19 and N-S_ε0.5 scenarios were calculated by the N-S equation, while others were given by the Darcy-Brinkman-Stokes equation. [c]Porosity-permeability relationship was determined by experimental measurements (Ehrenberg et al., 2006; Kaminskaite et al., 2019) and diffusion coefficients were calculated by Archie's law (Deng et al., 2016).

10.1029/2021WR030578

Table 3
Comparison of Effluent Concentrations of Calcium and Magnesium From Experiments and Different Models

Data concentration	Experiment (Ajo-Franklin et al., 2018)	Diffusion coefficient for the altered layer (m^2/s)	Reduced-dimension model (Deng et al., 2016)	Hybrid model (Molins et al., 2019)	Micro-continuum model (this work)
Calcium concentration/magnesium concentration (mol/L)	0.027/0.0087	2e−10	0.027/0.0094	0.031/0.0034	0.025/0.0033
		4e−11	0.021/0.0111	0.022/0.0049	0.014/0.0031

respectively, except that they neglect flow in the AL and the remainder of the rock matrix. This is accomplished by obtaining the velocity field using the Navier-Stokes equation in the fracture opening, rather than the DBS equation, which is also the approach used in Molins et al. (2019). The DBS_D_{low} scenario uses a lower diffusion coefficient for the AL, as also tested in Deng et al. (2016) and Molinset al. (2019), and the DBS_A_{high} scenario examines the effect of reactive surface area of dolomite.

4. Results and Discussion

Overall, results from the model presented here compare well with the experimental measurements (Table 3). Discrepancies are of the same order as those observed in previous modeling efforts (Deng et al., 2016; Molins et al., 2019; Table 3). Although assumptions, solution, and meshing approaches and codes are different, this indicates the ability of the model to capture the main processes in the fractured domain. A detailed description of the differences between results from our model and previous models result is presented in the Supporting Information.

4.1. Reactive Transport Regimes in the Fractured Dolostone

The results show spatial variations in velocities and concentrations that are strongly influenced by the structural heterogeneity (Figure 3). The velocity maps highlight a fast flow path on the right side of the fracture domain, whereas the left side is characterized by low velocities due to the presence of contact areas or small apertures. The downstream velocities are more uniform. Overall, regions with higher flow rates in the fracture correlate

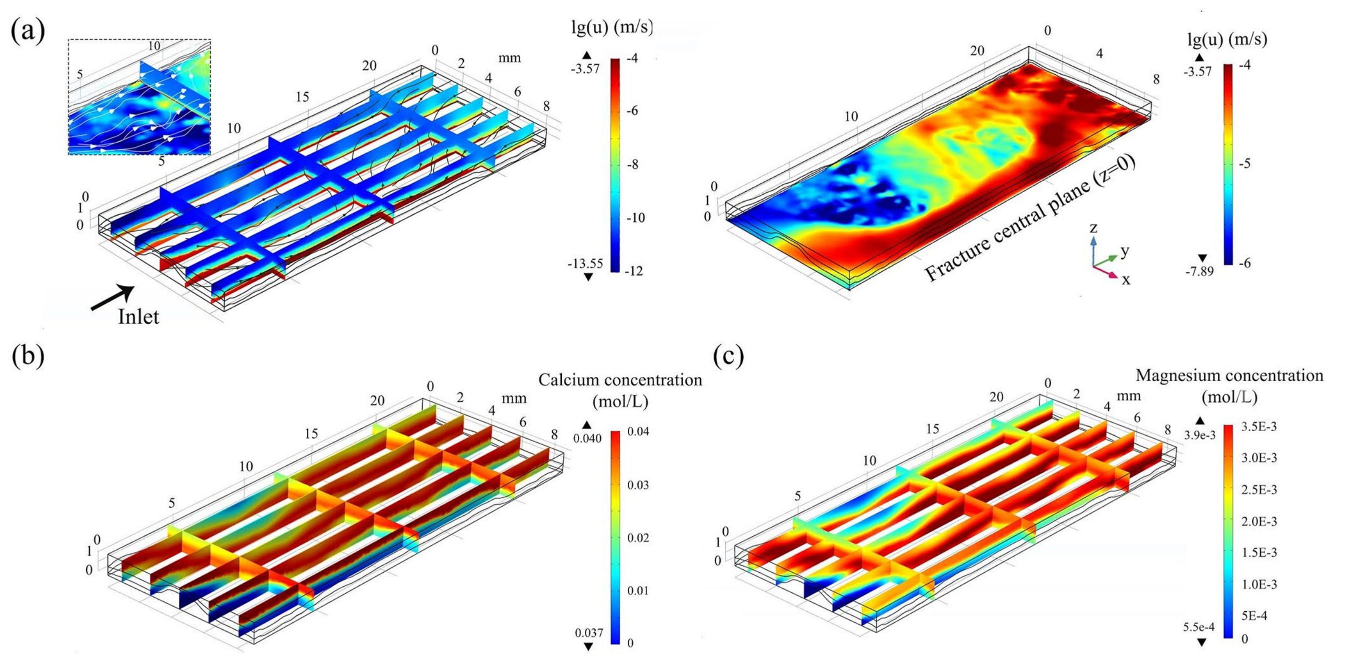

Figure 3. Visualizations of the (a) velocity field (including a close-up view of flow across the altered layer), (b) calcium concentrations, and (c) magnesium concentrations in the 3D domain.

to regions with lower concentrations of calcium (Ca) and magnesium (Mg). The average Darcy velocities are 4.52e−6, 1.02e−9, and 4.68e−11 m/s in the fracture, AL, and unreacted matrix, respectively. In the AL and matrix, the velocities are several orders of magnitude slower and is transport limited, and thus higher concentrations were observed. However, the streamlines shown in Figure 3a also highlight flow through the AL, especially at regions where there are contacts or fracture apertures are small. These features can have important implications regarding local reactions and thus subsequent geochemical evolution, as will be analyzed and discussed in detail below.

The dimensionless Péclet (Pe) and diffusive Damköhler (Da_{II}) numbers may be used to characterize the interplay between flow, transport, and reactions. For a given characteristic length L_c (fracture aperture here), Pe and Da are estimated in this work by $\frac{u_f L_c}{D}$ and $\frac{kL_c^2}{D}$, respectively. The reaction rate constant k (s^{-1}) is calculated as $k = \pi r_s/nL$, where S is the specific surface area, r_s is the intrinsic reaction rate without transport limitation, and n is the moles of mineral per unit volume, that is, $n = \frac{\rho_{\mathrm{mineral}} f_{\mathrm{mineral}}}{M_{\mathrm{mineral}}}$. While the cut-off value for determining the dominance of diffusion versus advection is not necessarily 1, the evident increase in the $\log_{10}(Pe)$ values (from −5.64 to −1.78 and 1.42) suggests a decreasing control of the diffusive transport process moving from the matrix to the AL and to the fracture. The rate-limiting process for different mineral reactions differs in different zones. With the shielding effect of the AL, calcite dissolution in the matrix is diffusion-limited as also confirmed by the large $\mathrm{Da_{II}}$, 365. For dolomite, $\mathrm{Da_{II}}$ is 0.389 and 0.006 in the matrix and AL, respectively, indicating an increased control of surface reaction. The advective Damköhler (Da_I) number, as given by Da_{II}/Pe, is on the order of 0.1 and suggests a more comparable control of surface reaction and advective transport on the reaction rate in the AL.

Figure 4 shows two-dimensional (2D) cross sections of concentrations and reaction rates, to illustrate the chemical gradients across the three regions. Two representative slices were chosen. One cuts through the flow channel in the fracture (yz_Slice 2), and the other is located in the region where the fracture is tight (yz_Slice 1). Generally, the dissolution of calcite only happens at the interface between the unreacted matrix and the AL, and Ca concentration is much higher in the matrix. The narrow region of reactions and the low effective reaction rates, even with a large surface area, highlight the diffusion limitations on calcite dissolution. For dolomite, dissolution is localized within the AL with higher reaction rate closer to the fracture surface. The Mg concentration is comparable in the AL and the unreacted matrix, and is much lower in the fracture. Comparison of the chemical profiles in the two locations provide further insights regarding the impacts of fracture geometry heterogeneity. In particular, fast flow in the wide fracture (yz_Slice 2) corresponds to lower Ca and Mg concentrations in the matrix, which enhances the geochemical disequilibrium and thus the extent of dissolution for both calcite and dolomite. This indicates a positive feedback, that is, porous media close to larger fracture apertures are the preferred regions of geochemical reactions (Fazeli et al., 2019). It leads to the increasingly heterogeneous pattern in both matrix alteration induced by calcite dissolution, and AL development and fracture enlargement from dolomite dissolution (Al-Khulaifi et al., 2019).

4.2. The Impacts of AL Processes on Bulk Chemistry

Variations in reactive transport parameters of the AL were implemented in different scenarios to examine the controlling factor on dissolution behaviors of both calcite and dolomite. Probability distribution functions (PDFs) of velocities (Figure 5) provide an overview of changes in the advective transport. For the DBS simulations, three peaks are observed, representing the flow in the matrix, the AL, and the fracture, from left to right. The velocity distributions in the fracture in the N-S simulations (Figure 5) overlap with the fracture peak of the DBS velocity distribution. The higher porosity and permeability used in the DBS_ε0.5 scenario enhanced both advective and diffusive transport in the AL compared to the base case. The middle peak shifts significantly to the right as flow velocities in the AL are two orders of magnitude higher than those in the base case. The shape of PDFs also shows a smoother transition of velocity distributions near fracture surfaces. As a result, the relative contribution of advective transport in the AL increases, which is confirmed by a more than one order of magnitude increase in the Pe number ($\log_{10}(Pe)$ is −0.55). This implies that advection in the AL could play a more important role when the AL is more permeable.

Table 4 summarizes the effluent concentrations and bulk reaction rates from all six scenarios to quantify the impacts of different properties of the AL on bulk chemistry. Compared to the base case, the DBS_D_{low} scenario results in the largest decrease (73%) in the effluent Ca concentration and the DBS_A_{high} scenario leads to the

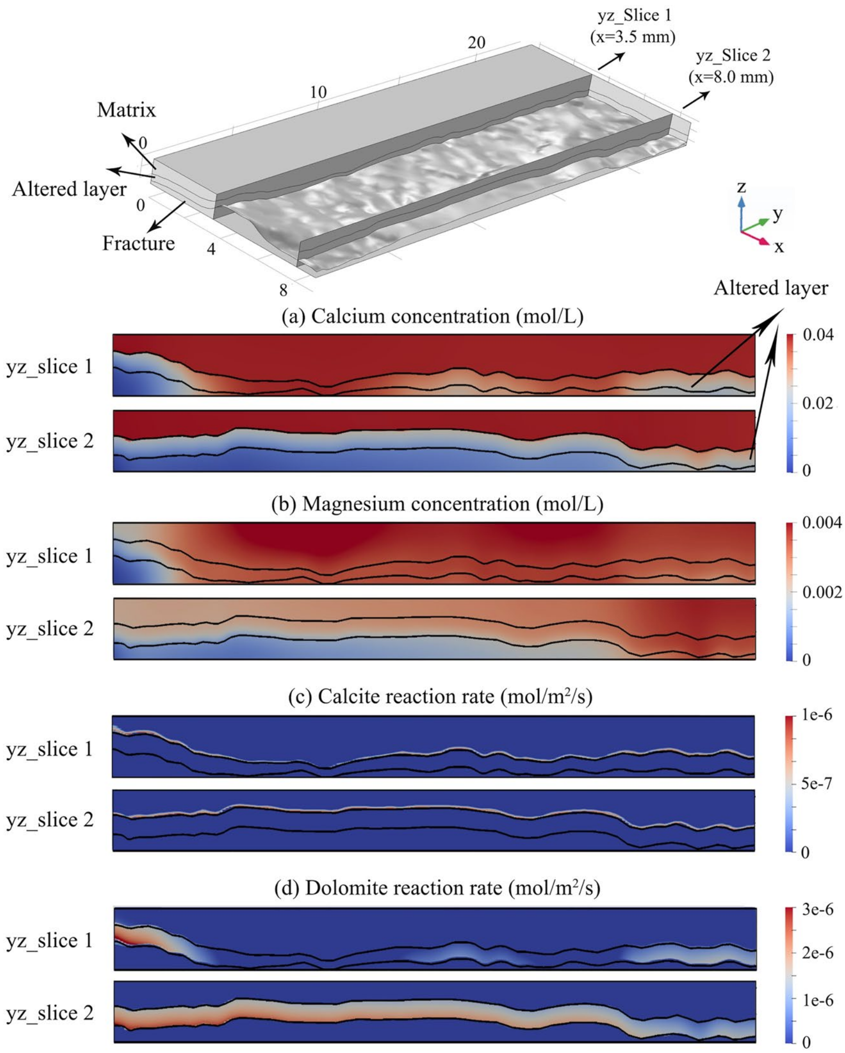

Figure 4. Chemical properties, that is, concentrations of (a) calcium and (b) magnesium, and reaction rates of (c) calcite and (d) dolomite from specific 2D profiles, yz_Slice 1 and yz_Slice 2 cut through the slow flow region and the fast flow path along the flow direction, respectively. The z-axis of slices is exaggerated by a factor of 2 to highlight variations through different domains.

largest increase (113%) in the effluent Mg concentration. The significant increase in the effluent Mg concentration observed in the DBS_A_{high} scenario supports that the reactive surface area for dolomite used here which is sourced from the literature is low compared to the dolostone core.

As a result of the enhanced advection and diffusion across the AL in the DBS_ε0.5 scenario, overall dissolution of both calcite and dolomite are increased compared to the DBS_ε0.19 scenario. The effluent concentrations of Ca and Mg increase by 21% and 6%, respectively. Calcite dissolution increases with diffusivity as expected, but the level of the increase depends on advection in the AL. The increase in calcite dissolution when there is no advection in the AL (N-S_ε0.19 and N-S_ε0.5 scenarios) is 22%, which is less than the extent of increase when there is advection in the AL (DBS_ε0.19 and DBS_D_{low}), that is, 73%. For dolomite, higher diffusivity results in slightly less dissolution without advection. It indicates that the impact of the pH-dependent reaction pathway of dolomite is important as more calcite dissolution results in more buffering of the pH and lower kinetic rate for dolomite. With advection, however, dolomite dissolution increases with the diffusivity due to the improved transport. The bulk chemistry results from N-S_ε0.19 and N-S_ε0.5 scenarios are comparable to the ones using the DBS velocity field. The lack of difference in the bulk chemistry with and without considering advection in the AL is a result of the spatial rearrangement of reactions, rather than an indicator that advection in the AL is

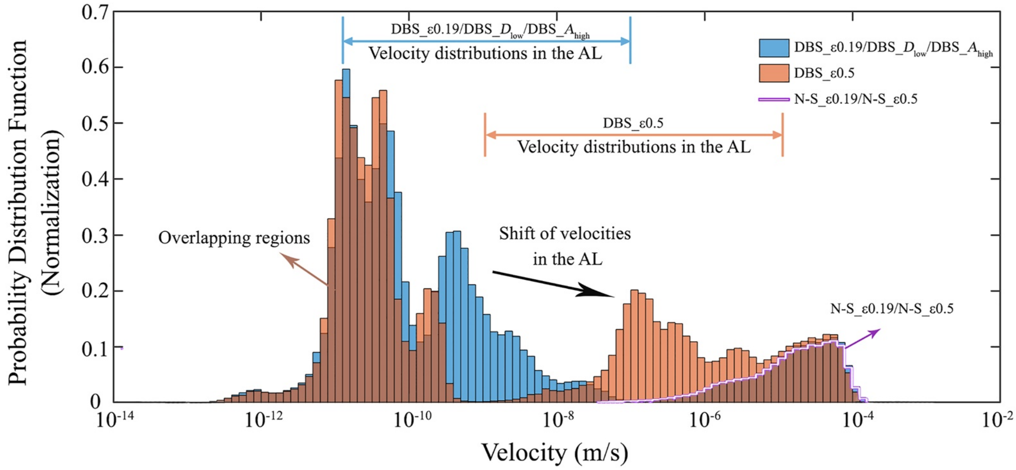

Figure 5. Comparison of probability distribution functions (PDFs) of velocities from six scenarios, grouping into three pairs. For N-S_ε0.19 and N-S_ε0.5 scenarios, the PDFs are calculated with zero values for the porous media while flow is only resolved in the fracture.

unimportant. As will be shown in Section 4.3, consideration of the advection in the AL results in spatial variations in local reaction rates.

In all the cases, bulk reaction rates of both calcite and dolomite are several orders of magnitude lower than the intrinsic reaction rates. Moreover, the effective reaction rates of dolomite are higher than those of calcite even though the intrinsic reaction rates are expected to be one order of magnitude lower. A similar phenomenon was observed in the previous experimental work (Al-Khulaifi et al., 2017) indicating limited calcite dissolution due to shielding provided by the remaining dolomite. These observations suggest that the slow-reacting mineral, that is, dolomite, can play a more important role in the later-stage alteration of fractured media when the AL becomes well established in a multi-mineral system, since the dissolution of the remaining minerals in the AL will control the subsequent fracture enlargement in addition to the AL development.

Overall, variations in the bulk reaction rates are similar to those in effluent concentrations, except for the case with the high surface area. In the DBS_A_{high} scenario, the higher reactive surface area results in more dolomite dissolution—with a significant increase (113%) in the effluent Mg concentration—but a lower surface area normalized rate due to more accumulation of the reaction products. This implies that dolomite reaction rates in the AL are strongly influenced by the surface reaction. Calcite dissolution is limited by both inefficient transport and exhaustion of fresh reactants. The net effect of the reduced calcite dissolution rate and increased dolomite dissolution rate is a higher effluent Ca concentration in this scenario.

Table 4
Comparison of Effluent Concentrations and Bulk Reaction Rates From Different Scenarios

		Transport metrics				Reaction metrics
	Base case	Advection-diffusion effects	Advection effects		Diffusion effects	Surface area effects
Effluent concentrations (mol/L)	DBS_ε0.19 scenario (base case)	DBS_ε0.5 scenario	N-S_ε0.19 scenario	N-S_ε0.5 scenario	DBS_D_{low} scenario	DBS_A_{high} scenario
Calcium	0.0248	0.0300	0.0249	0.0306	0.0143	0.0253
Magnesium	0.00326	0.00345	0.00325	0.00322	0.00309	0.00694
Bulk reaction rate (mol/m^2/s)						
Calcite	5.05e−9	6.29e−9	5.32e−9	6.74e−9	2.62e−9	4.30e−9
Dolomite	2.65e−7	2.81e−7	2.63e−7	2.60e−7	2.53e−7	1.12e−7

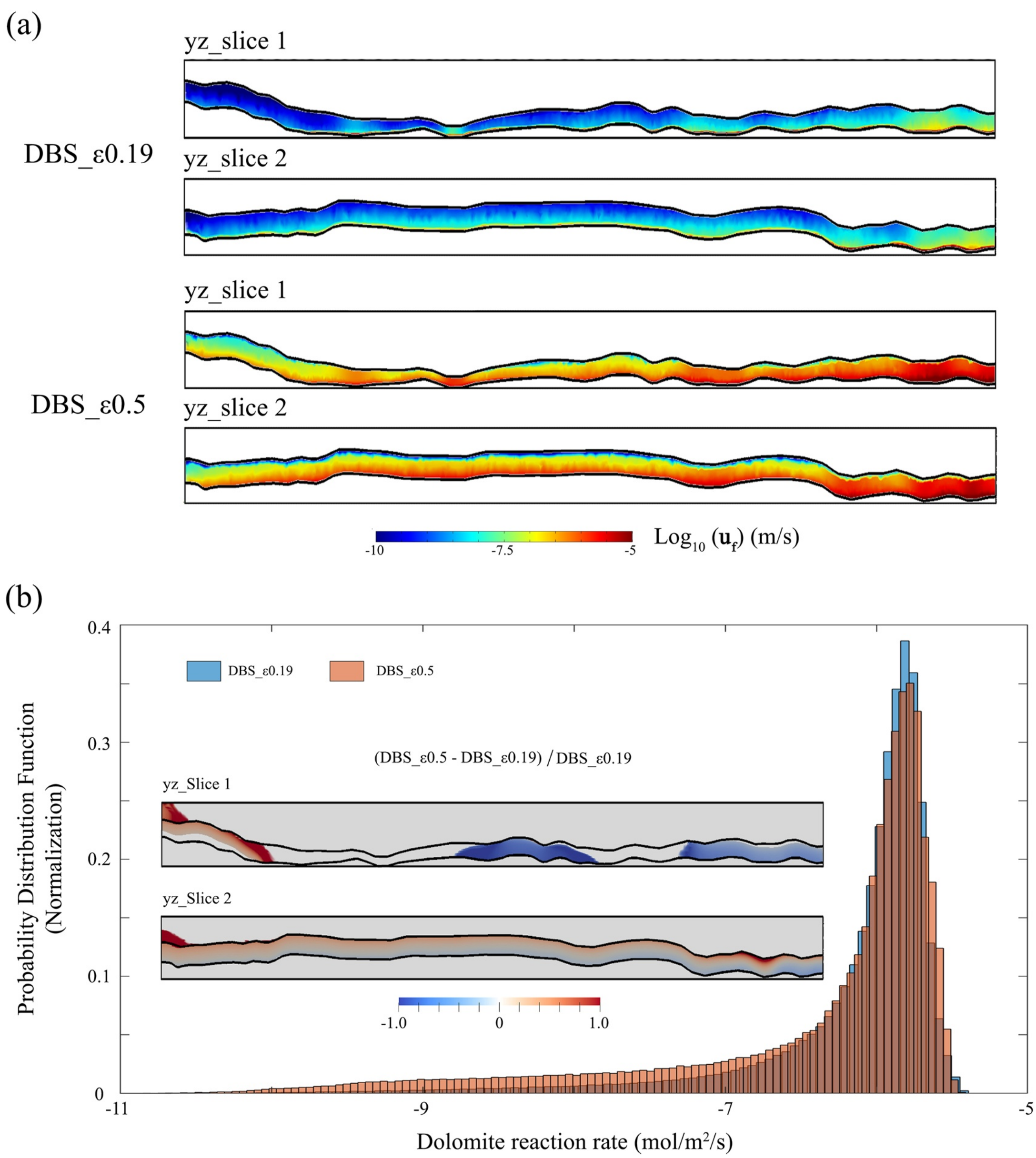

Figure 6. Comparisons of (a) spatial distributions of velocity (u_f) and (b) probability distribution functions and relative differences of dolomite reaction rates for DBS_ ε0.19 and DBS_ε0.5 scenarios.

4.3. Spatial Differences in Reactions and Implications on Alteration

Because long-term alteration of the fractured media is dictated by local reaction rates whereas bulk chemistry is an integrated outcome of the system dynamics, here, we report a detailed comparison of different scenarios focusing on local dolomite reaction rates, to better understand the long-term implications of different AL processes.

4.3.1. Coupled Effects of Advective and Diffusive Transport on Spatial Dynamics

Comparison of the DBS_ε0.19 and DBS_ε0.5 scenarios highlight the coupled effects of advective and diffusive transport. Spatial distributions of velocity (Figure 6a) illustrate the non-uniform flow in the AL with higher velocities in regions close to the fracture surface for both DBS_ε0.19 and DBS_ε0.5 scenarios, especially where fracture apertures are smaller. The PDF of dolomite dissolution rates (Figure 6b) of the DBS_ε0.5 scenario shows a flatter and a slightly more uniform distribution, that is, there are more grid cells with high or low reaction rates.

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The 2D profiles indicate that the increase in dolomite dissolution resulting from the increased advective and diffusive transport in the AL primarily occur in the regions with large apertures (e.g., yz_Slice 2 in Figure 6b). In the wider fracture region, dolomite reaction rate increase (more than 100% difference) is more evident close to the interface between the unreacted matrix and the AL, implying a stronger reaction front into the matrix, whereas dolomite reaction rate is reduced at the fracture surface. This means that the gradient in dolomite dissolution perpendicular to the fracture plane (as shown in Figure 4d) is more homogenized in these regions. In comparison, in regions with small apertures (yz_Slice 1 in Figure 6b), dolomite dissolution rate is mostly reduced except for zones close to the inlet before the channel flow is well developed. These observations imply that in rocks with higher fraction of fast-reacting minerals as in the DBS_ε0.5 scenario, the spatial heterogeneity of AL may be enhanced in the fracture plane, while staying more uniform vertically, which is an important assumption for using the analytical representation of diffusion limited reaction rate and as also assumed in the initial setup of the model here. The changes of aperture enlargement, however, will be more complex as even though the dissolution is dampened at the fracture surface, less remaining mineral is to be removed in this case.

4.3.2. Effects of Advective Transport on Spatial Dynamics

Comparisons of the N-S_ε0.19 and N-S_ε0.5 scenarios with the DBS_ε0.19 and DBS_ε0.5 scenarios provide a direct measure of the effect of advection within the AL. Spatial variations in the changes in dolomite dissolution rate are highlighted in Figure 7, and are especially evident (up to 20% difference in local reaction rate) in the case with large porosity in the AL as expected (i.e., DBS_ε0.5 and N-S_ε0.5). For the ε0.19 scenarios, advection in the AL increases dissolution of dolomite in the AL, mostly in the small aperture regions. In the ε0.50 simulations, advection increases dissolution of dolomite in the AL particularly in the large aperture regions; whereas the dissolution is enhanced in the upstream outside of the contact points and is reduced in the downstream in the small aperture regions.

These spatial variations even though limited, can still propagate throughout time. In a dynamic simulations in which the geometry/porosity is updated, spatial variations can be amplified following the positive feedback between reactions and transport, that is, the more reactive regions have more significant porosity increase and thus permeability and diffusivity enhancement, which in turn accelerate local reactions (Deng & Spycher, 2019; Fazeli et al., 2018; Garcia-Rios et al., 2017). As shown in Figure 4d, dolomite dissolution is faster in the wide aperture regions. The spatially varied enhancement in dolomite dissolution rate is likely to compound the spatial heterogeneity.

4.3.3. Effects of Diffusive Transport on Spatial Dynamics

Comparisons between the N-S_ ε0.19 and N-S_ ε0.5 scenarios (i.e., Group 1), and between the DBS_ε0.19 and DBS_D_{low} (i.e., Group 2) isolate the effects of diffusivity in the AL. In both groups, the diffusivity differs by a factor of 5. Figure 8 shows the probability and spatial distribution differences of dolomite reaction rates from two groups. Overall, lower diffusivity in the AL increased dolomite dissolution rates (the relative difference can be more than 100%) close to the fracture- AL interface and reduced the rates close to the interface between the AL and the unreacted matrix. As a result, dolomite dissolution rate has a larger gradient across the AL and the high dissolution rates are rearranged into a narrower zone.

In addition, the regions with slow velocities in the fracture (yz_Slice 1) are strongly affected by the diffusivity compared to the region of fast flow paths (yz_Slice 2). This is especially evident when advection is considered (i.e., Group 2), as also highlighted by the red circle in Figure 8b. Without advection, local reaction rate is solely determined by the interplay between diffusion and surface reaction, whereas with advection, local reaction rate may be controlled by advection and surface reaction interactions instead, as also discussed in Section 4.1. As such, those regions with higher advection are not affected by the diffusivity reduction, and the reduced reactions caused by stronger diffusion limitations in other places can preserve higher thermodynamic driving force for surface reaction and thus higher local reaction rate in these regions.

Overall, the PDF of the reaction rate in the case without considering advection in the AL is narrower with the lower diffusivity. In contrast, with advection the PDF spreads over a wider range and is flatter, indicating a more heterogeneous spatial distribution when the diffusivity is lower. While the lower diffusivity in the AL highly impedes the alteration of unreacted matrix induced by calcite dissolution as observed in previous experiments (Al-Khulaifi et al., 2019; Deng et al., 2017; Ellis et al., 2013), the increase in dolomite dissolution close to the

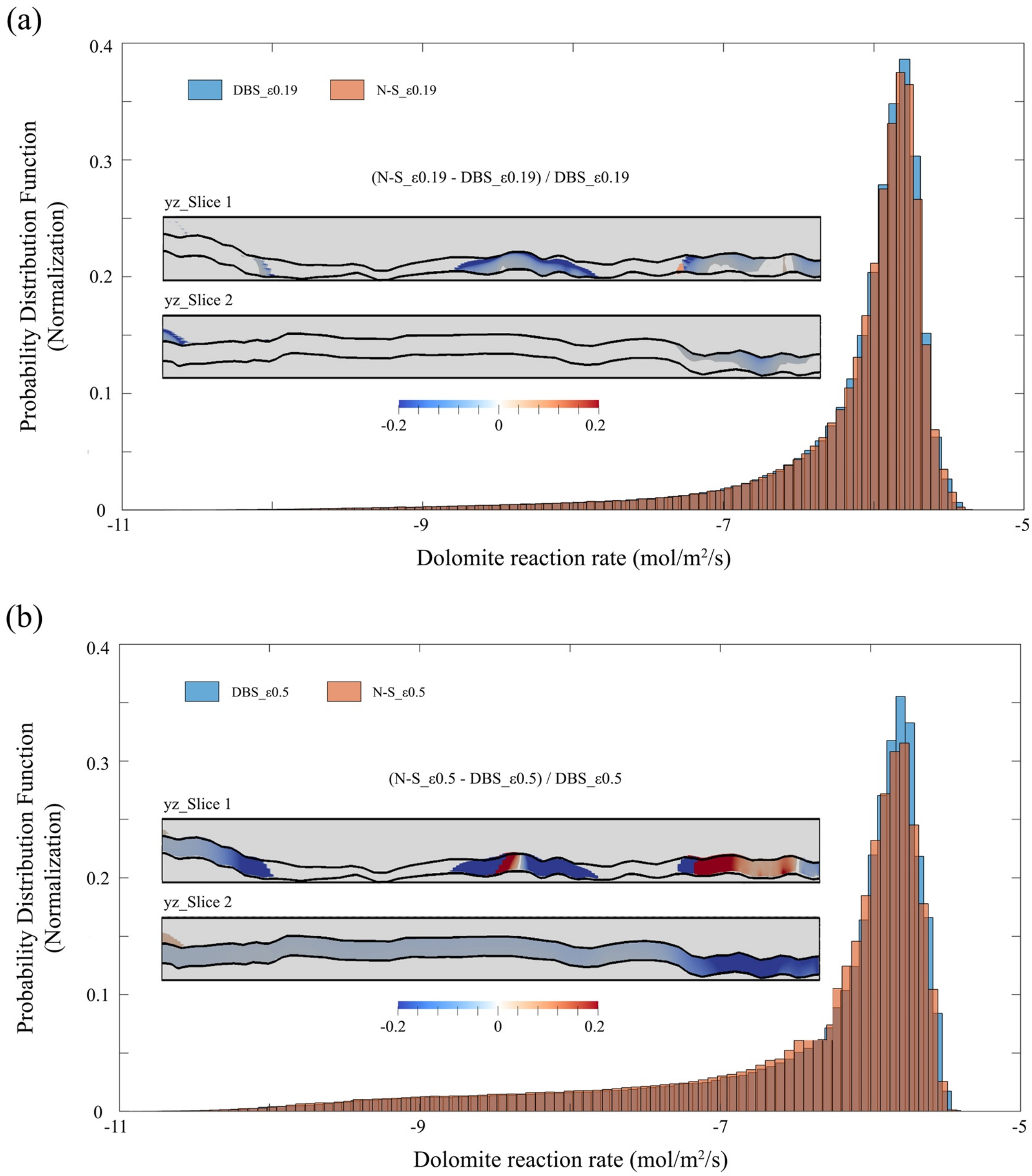

Figure 7. Comparison of dolomite reaction rates by probability distribution functions and relative differences for (a) DBS_ ε0.19 and N-S_ε0.19 and (b) DBS_ε0.5 and N-S_ε0.5.

fracture-AL interface—the level of which depends on whether advection is accounted for—will enhance fracture enlargement.

4.3.4. Effects of Reactivity on Spatial Dynamics

The DBS_A_{high} scenario with a higher reactive surface area of dolomite was performed to examine the effects of surface reaction. However, the increase in surface area and that in the extent of dolomite dissolution are not proportional, confirming the mixture of transport-controlled reaction and surface-controlled reaction (Section 4.1). The lower peak of PDF for DBS_A_{high} compared to DBS_ε0.19 is caused by the presence of zero dolomite dissolution rate in more grid cells. Relative differences (Figure 9) in dolomite dissolution between DBS_A_{high} and DBS_ε0.19 provide an indicator of the rate-limiting regimes. The red color corresponds to regions with significant rate increase where the reactions are more likely surface-controlled, whereas the blue indicates the regions of stronger transport limitation. Comparison of different slices highlights that close to larger fracture apertures,

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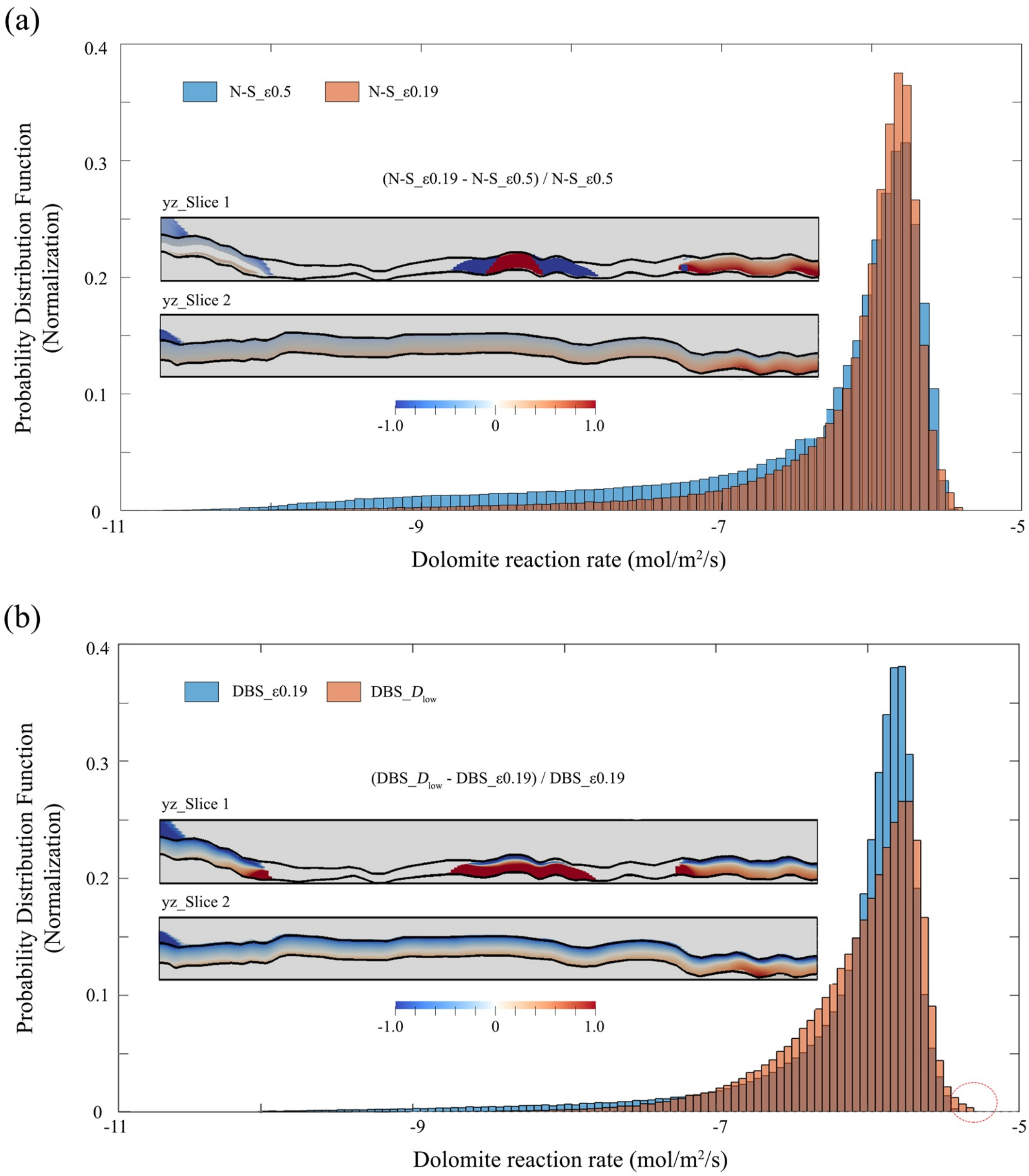

Figure 8. Comparison of dolomite reaction rates by probability distribution functions and relative differences for (a) N-S_ε0.19 and N-S_ε0.5 and (b) DBS_ε0.19 and DBS_D_{low}. The red color circle highlights the larger reaction rates from DBS_D_{low} scenario.

reactant delivery and mixing are more effective and the reactions are primarily controlled by surface reactions. It also shows that reactions mainly happen close to the inlet and decrease along the flow path, which gives rise to a non-uniform evolution pattern of fractured media.

5. Conclusions and Implications

In this study, a DBS-based micro-continuum reactive transport model was developed by coupling the COMSOL multiphysics code and CrunchFlow. Using this modeling framework, we investigated the reactive transport processes in a fractured domain including a pore-scale component and a Darcy-scale component. The domain was reconstructed from a fractured dolostone after reacting with CO_2-acidified fluid, which resulted in a porous AL bordering the fracture where calcite is depleted and the dolomite remains. In our simulations, flow, diffusion, and reactions were explicitly considered in the open fracture, the bordering porous AL, and the unreacted matrix,

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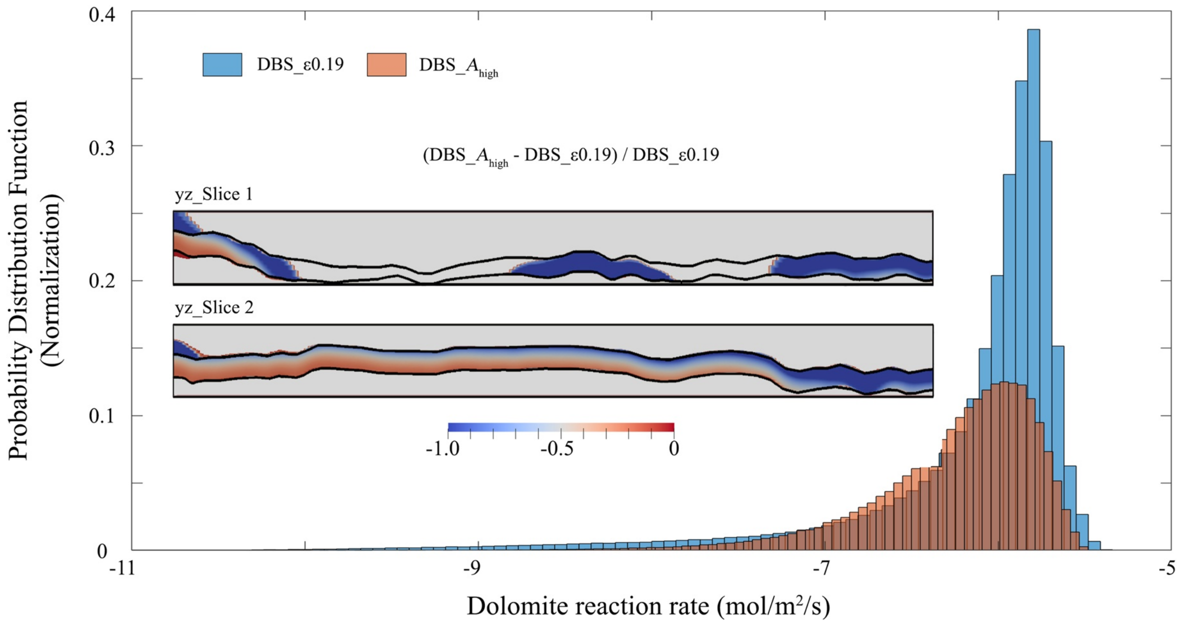

Figure 9. Comparison of dolomite reaction rates by probability distribution functions and relative differences for DBS_ε0.19 and DBS_A_{high}.

allowing accurate depiction of mass transfer within and across domains of contrast properties. By examining the bulk and local reactions at steady state for different simulation scenarios, we were able to, for the first time, investigate rigorously the impacts of the physical-chemical properties within the AL on reactive transport in the fractured media.

Our simulation results show good agreement with previous experimental and modeling studies of the same system. Calcite dissolution occurs primarily at the interface between the unreacted matrix and the AL, and its rate is strongly diffusion limited. Significant reduction in overall calcite dissolution (73%) was observed when the diffusivity is reduced by a factor of 5. In contrast, it is not significantly affected by advection in the AL. In this system, the reaction rate of calcite (i.e., the so-called fast-reacting minerals) can be slower than that of dolomite (i.e., the nominally slow-reacting minerals), indicating a critical role of the remaining mineral in the subsequent alteration of fractured media. Dolomite dissolution exhibits complex spatial variations in response to the changes in advective flow, diffusion, and reactive surface area in the AL. Advective flow provides an additional transport mechanism and tends to enhance dolomite dissolution within the AL in certain regions (by ∼20%), especially when the porosity and thus permeability of the AL is high. The differences are also more evident in the small aperture regions. A 7% increase in the overall dolomite dissolution rate was predicted when advective flow is considered in an AL with porosity of 50%. The difference in the spatial distribution of dolomite reaction rate due to presence of the advective flow suggests that the evolution of the fracture aperture and AL may differ from that which ignores advective flow, as advective flow tends to increase spatial heterogeneity. In addition, when advective flow in the AL is considered, dolomite dissolution responds differently to the changes in the diffusivity of the AL. Overall dolomite dissolution increases as diffusivity decreases when advection in the AL is not considered, whereas the overall reaction decreases as diffusivity decreases when advection is considered. For a lower diffusivity in the AL, the dissolution rate of dolomite is enhanced at the fracture surface, which means enhanced fracture enlargement, and reduced at the matrix-AL interface, and the extent is more evident when advection is considered.

While bulk chemistry provides important insights on the overall behavior, spatial variations in local reactions are key for controlling long-term dynamics in heterogeneous fractured media. Overall, our simulations showed that even though advective flow in the open fracture is dominant, advective flow in the AL can still be important in controlling reactive transport in the system and thus the evolution of the fracture and the AL, especially when the AL is permeable or the fracture apertures are small. In these cases, explicit consideration of continuity of fluid fluxes and pressures at the interface between fractures and the bordering porous media is needed for accurate prediction. Moreover, reactions of the nominally slow-reacting minerals should be resolved in reactive transport simulations as they directly affect the long-term alteration of fractures. While not considered in this study, time-evolving domains resulting from geochemical reactions can also be simulated in the COMSOL-CrunchFlow framework. Our future work will focus on capturing the dynamics in geometries and hydraulic properties of complex fractured porous media and the feedback with geochemical reactions. Such modeling efforts will be especially useful for co-design of simulations and experiments in 3D to fill gaps in theoretical understanding and observations of intimately coupled processes that control the evolution of fractured porous media.

Data Availability Statement

Input files for the multiscale modeling of the fractured dolostone, DOI: https://doi.org/10.5281/zenodo.4594959.

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